



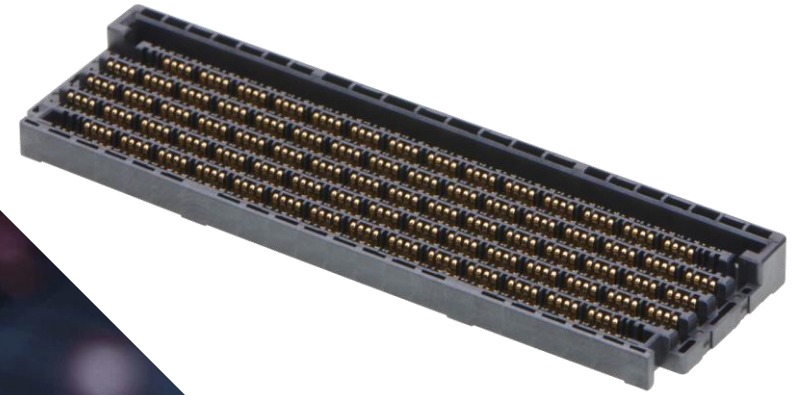
Customer Presentation

MIRROR MEZZ CONNECTORS

PRODUCT NOTICE

PRESENTER:

DATE:



Mirror Mezz Connectors

The Mirror Mezz family of hermaphroditic mezzanine connectors delivers superior signal integrity (SI) performance and data speeds up to 224 Gbps to address ever-increasing data transmission requirements for telecommunications, networking and other high-density applications, including Open Compute Project (OCP)-compliant systems.

Key Product Information

Category: Board-to-board mezzanine connectors

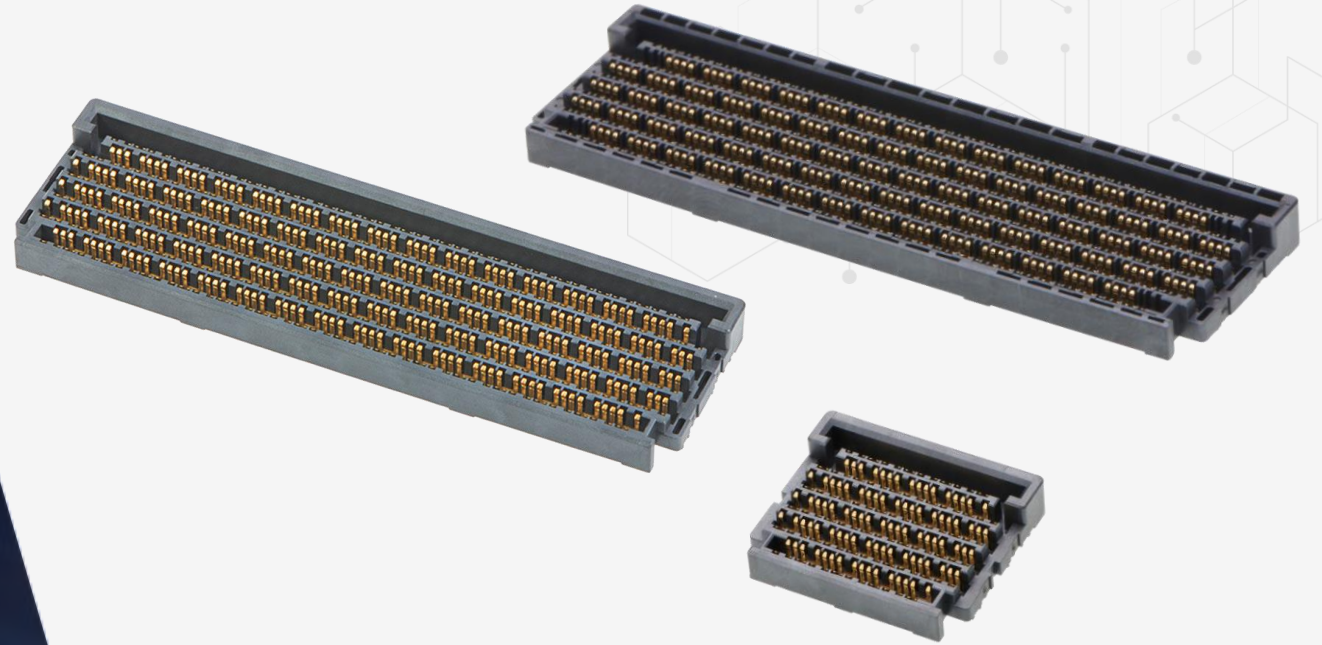
Data Rate: Up to 224 Gbps

Pitch: 4.00 by 1.50mm differential pair pitch

Stack Height: 5.00, 8.00, 11.00mm

Pin Count: Up to 270 differential pairs

Standards: OCP (Mirror Mezz, Mirror Mezz Pro)



[View Product
Landing Page](#)

[Download Datasheet](#)

Series

202828, 204358, 206306, 209141, 209311

Mirror Mezz 112 Gbps Connectors

218910, 218916

Mirror Mezz Pro 112 Gbps Connectors

224048

Mirror Mezz Enhanced 224 Gbps Connectors

Vital Product Information

Mirror Mezz Connectors

What makes this product different from the competition?

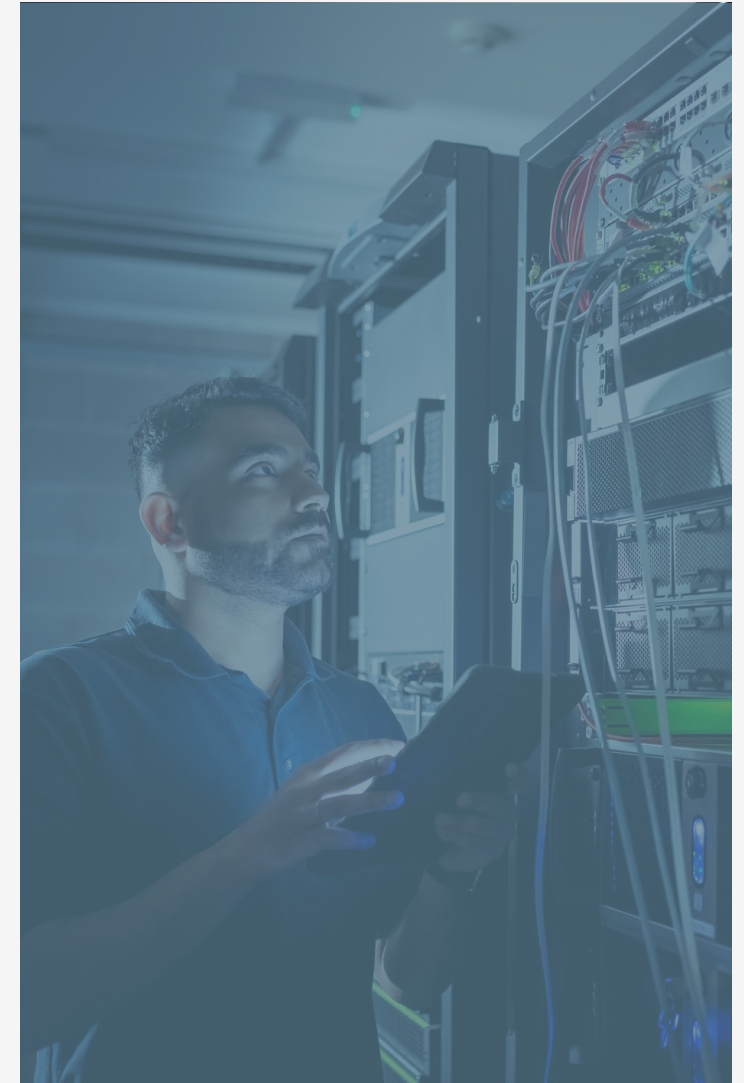
A stackable hermaphroditic design with an available OCP-standard 15x11 compatible footprint and up to 224 Gbps data transfer rate makes Mirror Mezz Connectors a unique combination of compactness, performance and design flexibility.

How does this product/solution create value for our customers?

Complementary pairing of identical (hermaphroditic) connectors lowers tooling and inventory costs while improving operational efficiency and optimizing space in height-constrained applications. Ball grid array (BGA)-attach mounting provides consistent solder ball profiles, leading to improved signal integrity (SI) for the entire connector system.

What is the Molex advantage?

Global manufacturing capability, robust engineering support, OCP-recommended connectivity products, and complementary Mirror Mezz-compatible solutions such as Copper flex and high-performance cables make Molex a unique partner in meeting the challenges of next-generation data center architecture.



Product Overview

Mirror Mezz Connectors

Mirror Mezz Connectors

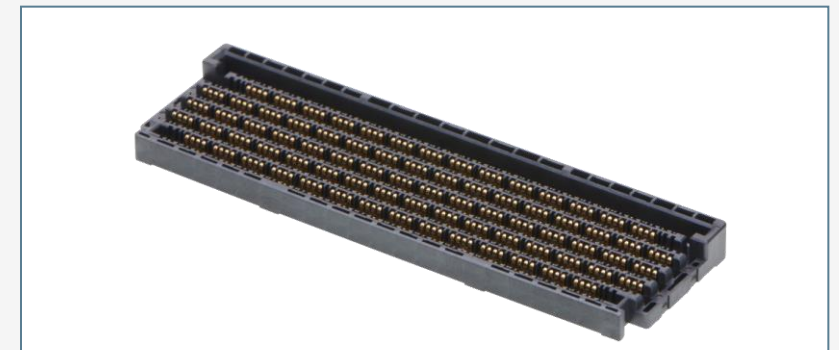
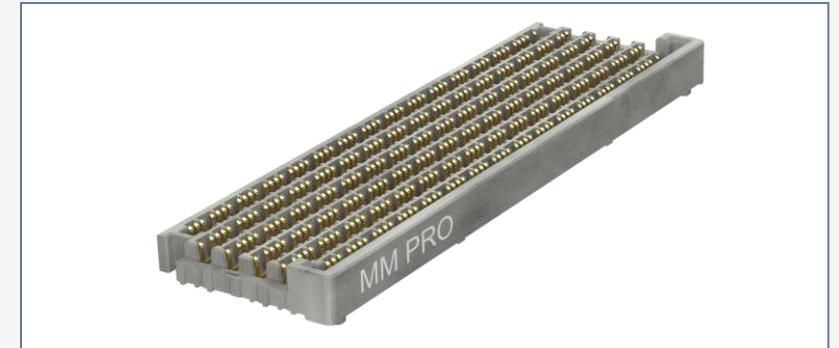
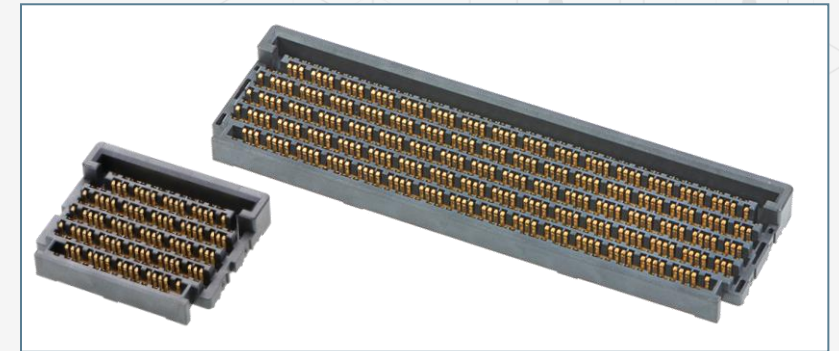
Recommended by the Open Compute Project (OCP) for board-to-board mezzanine interconnect applications, Mirror Mezz Connectors deliver superior signal integrity and up to 112 Gbps performance in a robust, hermaphroditic form factor and a variety of circuit sizes.

Mirror Mezz Pro Connectors

Mirror Mezz Pro Connectors provide 112 Gbps performance in an OCP-recommended form factor, enabling three stack height options with up to 270 differential pairs in a high-density layout with a robust “stubless” contact interface and blind-mating capability for easy assembly.

Mirror Mezz Enhanced 224 Gbps Connectors

As increased use of AI drives demands for faster speeds, the Mirror Mezz Enhanced Connector answers the challenge with data rates up to 224 Gbps PAM-4 for next-generation applications. Mirror Mezz Enhanced Connectors include the high SI, robust construction and hermaphroditic design of other Mirror Mezz Connectors, supporting design flexibility and system reliability.



Markets and Applications

Mirror Mezz Connectors



Storage Systems



Networking Equipment



Electronic Control Units

SERVER AND STORAGE

- Networking equipment
- Storage systems
- Server racks
- Open Accelerator Infrastructure (OAI) equipment

TELECOMMUNICATIONS

- Telecommunications infrastructure
- Networking equipment

AUTOMOTIVE

- Electronic control units
- Domain control units
- Zonal control units

Frequently Asked Questions

Mirror Mezz Connectors

What are the differences between Mirror Mezz Enhanced Connectors and other Mirror Mezz solutions?

Mirror Mezz Enhanced Connectors support faster signal speeds of up to 224 Gbps and have improved SI performance when compared with Mirror Mezz and Mirror Mezz Pro Connectors.

Which Mirror Mezz solution is right for my application?

Mirror Mezz, Mirror Mezz Pro and Mirror Mezz Enhanced Connectors offer excellent data speeds and SI to meet next-generation data center requirements. Molex offers a wide variety of circuit counts for different applications. Please contact us for details.

How many stack height options does the Mirror Mezz Enhanced Connector offer?

All Mirror Mezz Connector variants are available with three different stack heights of 5.00, 8.00 and 11.00mm through the combination of 2.50- and 5.50mm-high connectors.

Do Mirror Mezz Connectors comply with OCP design recommendations?

Mirror Mezz and Mirror Mezz Pro Connectors are recommended by OCP for mezzanine board-to-board applications.

Solving Industry Challenges

Mirror Mezz Connectors



Industry Need	Industry Challenge	Industry Solution	Anticipated Results
Faster data rates and excellent SI	As Internet of Things (IoT) and artificial intelligence (AI) applications drive demand for faster data rates and higher SI, data centers must upgrade their capabilities in a cost-effective manner.	Mirror Mezz Connectors deliver data rates up to 224 Gbps, with a common footprint and OCP-compliant designs that help future-proof data center architecture and streamline upgrades.	Customers can accommodate demands for faster speeds and improved SI performance while minimizing system redesign work and supporting scalability.
Compactness and density	PCB real estate is becoming increasingly scarce as customers look to pack more functionality into the same area to meet the requirements of a range of applications.	Mirror Mezz Connectors feature up to 270 differential pairs (DPs) and 107 to 115 DPs per square inch, a much higher density than the industry average, leading to significant space savings and more cost-effective use of PCB real estate.	Space savings on the PCB translate into significantly lower overall applied costs for any high-volume environment over an extended period of time.
Cost savings	High-density and enterprise server applications need fewer components, so they can reduce costs and increase operational efficiency.	The hermaphroditic Mirror Mezz Connector is designed to reduce tooling, inventory and operational costs while increasing time and process efficiencies for customers. The BGA-attach termination also streamlines assembly work, while the common footprint simplifies design work.	Customers using Mirror Mezz Connectors have fewer interconnect components to manage in their inventory. Cost and time savings improve overall productivity while reducing supply chain risks.

Product Advantages and Features

Mirror Mezz Connector Family

Reduces mis-mating and assembly errors through a robust shrouded housing design

Shrouding encapsulates the pin field, helping protect pins and offering blind-mate guidance to help eliminate mis-mating.

Lowers costs through hermaphroditic design

The self-mating design simplifies the bill of materials (BOM) and inventory management while minimizing tooling requirements.

Offers increased reliability and minimizes crosstalk between rows

Opposing beam support helps ensure improved performance through a 1.50mm row pitch.

Reduces costs through use of low-profile ball grid array (BGA) attachment

The BGA design in the Mirror Mezz product family offers greater cost savings, reduces lead times and simplifies the product matrix.

Simplifies assembly operations

The contact tip design offers fine alignment of 0.70mm in blind-mating scenarios, making mating operations easier and more reliable.

Helps to ensure constant electrical contact and improved reliability

Contact beam structure prevents vibrations and terminal lift to ensure two-point electrical contact, offers reliable normal force for harsh environments, and ensures sufficient engagement with 1.50mm of nominal contact wipe.

Key Specifications	
Connector Type	Hermaphroditic board-to-board mezzanine connector
Pitch	4.00 by 1.50mm
Connector Height	2.50 or 5.50mm
Stack Height	5.00, 8.00 or 11.00mm
Voltage	29.9V AC RMS
Current (max. per contact)	0.75A for 1 oz. Copper 1.2A for 2 oz. Copper

Product Advantages and Features

Mirror Mezz and Mirror Mezz Pro Connectors



Eases design challenges by maximizing PCB real estate

The high-density connector pin field includes up to 270 differential pairs.

Delivers reliable high-speed data transmission rates

Mirror Mezz and Mirror Mezz Pro Connectors support data rates up to 112 Gbps NRZ.

Offers relaxed tolerances and greater architectural flexibility with flex cable links

The design accommodates offsets between boards while flexible system components with flex cable links offer controlled channels and pinned grounds.

Improves design flexibility with 15x11 OCP-compliant design

The connector is recommended by OCP for 15x11 mezzanine board-to-board interconnect applications, ensuring interoperability for OCP-standard systems.

Optimizes use of space in height-constrained applications

Cross-mating or self-mating of 2.50- and 5.50mm-height connectors permits ultra-low and medium stack heights of 5.00, 8.00 or 11.00mm.

Key Specifications

Standard	OCP 15x11 mezzanine connector
Data Rate	Up to 112 Gbps
Differential Pairs	Up to 270
Color	Mirror Mezz – black Mirror Mezz Pro – gray
Mate Force (max. per pin)	0.35N
Operating Temperatures	-40 to +105°C

Product Advantages and Features

Mirror Mezz Enhanced Connectors

Delivers high-speed data transmission rates for next-generation applications

Mirror Mezz Enhanced Connectors offer 224 Gbps NRZ data speeds, supporting high-performance applications such as AI and machine learning.

Maximizes high-speed performance and clean routing out of the connector footprint

The precise arrangement of wide ground pins and electrically tuned signal contacts improves SI and helps balance the electrical field.

Improves SI performance and reliability

“Stubless” contact interfaces include two points of contact on each beam for greater reliability and reduce minimum stack height to 5.00mm.

Permits seamless upgrades to 224 Gbps speeds

Having the same footprint as Mirror Mezz and Mirror Mezz Pro Connectors helps reduce PCB redesign requirements.

Optimizes PCB real estate with higher differential pair (DP) density

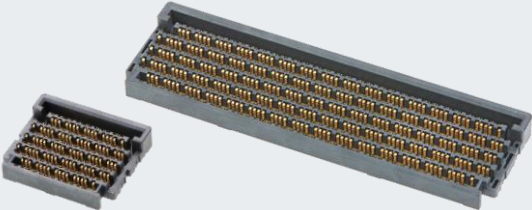
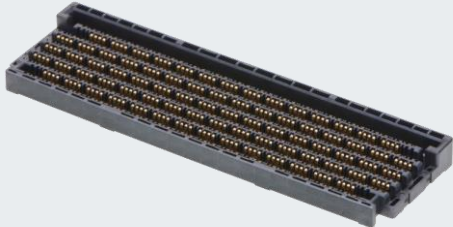
Mirror Mezz Enhanced features up to 270 full DPs and 24 single-ended pins, resulting in more DPs per square inch and enhancing the design’s versatility.



Key Specifications	
Data Rate	Up to 224 Gbps
Differential Pairs	Up to 270
Color	Black
Mate Force (max. per pin)	0.35N
Operating Temperatures	-40 to +85°C

Differentiation Across the Mirror Mezz Family

Mirror Mezz Connectors

	Mirror Mezz	Mirror Mezz Pro	Mirror Mezz Enhanced
Series Number	202828, 204358, 206306, 209141, 209311	218910 , 218916	224048
Data Rate	112 Gbps		224 Gbps
Pitch	4.00 by 1.50mm		
Current (max. per contact)	0.75A for 1 oz. Copper; 1.2A for 2 oz. Copper		
Mate Force (max. per pin)	0.35N		
Connector Height	2.50 or 5.50mm		
Differential Pairs	Up to 270		
OCP 15x11 Standard	Yes		No
Operating Temperatures	-40 to +105°C		-40 to +85°C
Product Image			

Product Specifications

Mirror Mezz Connectors

Reference Information

Packaging: Tape and reel
Mates With: 2.50- and 5.50mm-height connectors can self- or cross-mate
Designed in: Millimeters
RoHS: Yes
Halogen Free: Yes

Electrical

Voltage (max.): 29.9V AC RMS
Current (max. per contact):
Mirror Mezz, Mirror Mezz Pro: 1.0A
Mirror Mezz Enhanced: 0.75A
Low-Level Contact Resistance:
30 milliohms max. (initial)
10 milliohms max. (delta)
Dielectric Withstanding Voltage: 500V DC
Insulation Resistance: 1,000 Megohms
Impedance: 90 Ohms

Mechanical

Pitch: 4.00mm between differential pairs,
1.50mm between rows
Circuits: Up to 270 differential pairs
Durability (max.): 100 cycles
Mate Force (max. per pin):
Mirror Mezz, Mirror Mezz Pro: 0.35N
Mirror Mezz Enhanced: 0.5N
Gatherability: 1.20mm (Y-axis) and 1.00mm (X-axis)
Average Unmating Force (min.): 0.045N per pin

Physical

Housing: LCP
Contact: Copper Alloy
Plating:
Contact Area—Gold
Solder Tail Area—Tin
Underplating—Nickel
Operating Temperatures:
Mirror Mezz, Mirror Mezz Pro: -40 to +105°C
Mirror Mezz Enhanced: -40 to +85°C

Additional Resources

Web Overview Page	https://www.molex.com/en-us/products/connectors/board-to-board-connectors/mirror-mezz-mirror-mezz-pro-connectors
Datasheet	987652-0673.pdf (molex.com)
Global Product Manager	Wai Kiong Poon, CSBU, DSS



THANK YOU

creating connections for life

molex